

CREATIVE INSTITUTE DATA SCIENCE

Data Cleaning

Practical -4 fill missing value by mode for category data

Note:- mode only use for categorical missing value fill.

When we use mode .. data must be in random value...

Now import library file.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

load dataset

```
df=pd.read_csv('train.csv')
df.head()
```

see full columns in dataset

```
pd.set_option('display.max_columns',None)
pd.set_option('display.max_rows',None)
df.head()
```

check the shape of dataset.

```
df.shape
```

now find only categorical data from dataset

```
cat_data=df.select_dtypes(include='object')
cat_data.head()
```

find the shape of this dataset

```
cat_data.shape
```

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find the missing data in new dataset

```
cat_data.isnull().sum()
```

now find the percentage of missing categorical data

```
per_cat=cat_data.isnull().mean()*100  
per_cat
```

we have seen that there is some data whose have more than 20% of missing data we need to drop this columns

find the columns

```
col=per_cat[per_cat>20].keys()  
col
```

now drop this columns from dataset

```
cat_data.drop(columns=col,axis=1,inplace=True)
```

now find the dataset shape

```
cat_data.shape
```

now find the percentage of missing dataset

```
cat_data.isnull().mean()*100
```

find the missing variable name

```
var_miss=miss_per[miss_per>0].keys()  
var_miss
```

if we want to fill global constant for missing data for perticular columns

```
cat_data['MasVnrType'].fillna('Missing')
```

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there is method called mode for missing value impute.

We mostly use mode method for filling missing value

Now find mode of particular data.

```
cat_data['MasVnrType'].mode()
```

we clearly see that there is None value for most frequent data and get series of data.

Also we need to count the value of most frequent data..

```
cat_data['MasVnrType'].value_counts()
```

we clearly see that most frequent value is None value.

Now we need to fill value using mode method.

```
cat_data['MasVnrType'].fillna(cat_data['MasVnrType'].mode()[0])
```

we have only fill value for single variable .

```
for i in var_miss:  
    cat_data[i].fillna(cat_data[i].mode()[0],inplace=True)  
    print(i,'=',cat_data[i].mode()[0])
```

now we have filled missing value and also print the most frequent value
now check the fill value or not.

```
cat_data.isnull().sum()
```